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Fri 10 Jul 2026 20:08:27 - sadmlib_std_demo.py v03.28.00 - Library v4.69.4
Short description of program
sherlock.maison.ca - User: root - Umask: 0002 - Arch: x86_64
Redhat Linux v9.8 - Kernel 5.14.0-687.20.1.el9_8.x86_64 - SADMIN(S): /opt/sadmin
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[OK] Connection to sadmin database establish.

01) SADMIN Python Library functions available to developers

[001]	sa.get_release()	SADMIN Release Number (XX.XX)	2.1.2
[002]	sa.get_ostype()	OS Type (In Uppercase, LINUX, AIX, MACOS)	LINUX
[003]	sa.get_osversion()	Return O/S Version (Ex: 7.2, 6.5)	9.8
[004]	sa.get_osmajorversion()	Return O/S Major Version (Ex 7, 6)	9
[005]	sa.get_osminorversion()	Return O/S Minor Version (Ex 2, 3)	8
[006]	sa.get_osname()	O/S Distribution (In uppercase)	REDHAT
[007]	sa.get_oscodename()	O/S Project Code Name	N/A
[008]	sa.get_username()	Current User Name	root
[009]	sa.get_kernel_version()	Return O/S Running Kernel Version	5.14.0-687.20.1.el9_8.x86_64
[010]	sa.get_kernel_bitmode()	Return O/S Kernel Bit Mode (32 or 64)	64
[011]	sa.phostname	Return Host Name	sherlock
[012]	sa.get_host_ip()	Return Host Main IP Address	192.168.1.48
[013]	sa.get_domainname()	Return Host Domain Name	maison.ca
[014]	sa.get_fqdn()	Return Fully Qualified Domain Name	sherlock.maison.ca
[015]	sa.get_serial()	Return System serial Number	0
[016]	sa.get_epoch_time()	Return Current Epoch Time	1783728507
[017]	sa.epoch_to_date(1783728507)	Convert epoch time to date	2026.07.10 20:08:27
[018]	sa.date_to_epoch(wsdate)	Convert Date to epoch time	1783728507
	DATE1 (End): 2026.06.30 10:00:44	DATE2 (Start): 2026.06.30 09:50:03	
[019]	sa.elapse_time (DATE1,DATE2)	Elapse Time between two timestamps	00:10:41
[020]	sa.get_packagetype()	Return package type (rpm, deb, lpp, dmg)	rpm
[021]	sa.get_arch()	Get system architecture	x86_64
[022]	sa.lock_system(hostname)	Lock the specified hostname (Not FQDN)	True
[023]	sa.lock_status(hostname)	Check if host specified is lock	True
[024]	sa.unlock_system(hostname)	Unlock host specified, monitoring on	True
[025]	sa.lock_status(hostname)	Check if host specified is lock	False
[026]	sa.write_log('message'[,lf=true])	Write message to Screen, Log or Both	message
[027]	sa.write_err('message'[,lf=true])	Write message to Error and Std Log	message
[028]	sa.sendmail(addr, sub, body, attach)	Send email (body=textFile) & opt.attach	0=Success 1=Error
[029]	sa.show_version(sa.ver)	Show Program & Library version & Desc.	scriptName -v
[030]	sa.sleep(60,15)	Sleep 60 sec & update every 15 sec.	60...45...30...15...0
[031]	sa.trimfile(filename,nlines=500)	Keep last 500 lines of filename.	0=Success 1=Error
[032]	sa.get_nb_cpu	Return number of CPUs	2
[033]	sa.get_nb_core_per_socket()	Return number of core per socket	2
[034]	sa.get_nb_thread_per_core()	Return number of threads per core	1
[035]	sa.get_nb_socket()	Return number of socket	1
[036]	sa.silentremove('file')	Silent file delete (no msg, no err)	0

[037]	sa.touch_file('filename')	Create an empty file	0=Success 1=Error
[038]	sa.db_connect('sadmin')	Open connection to database	0=Success 1=Error
[039]	sa.db_close(conn,cur)	Close connection to database	0=Success 1=Error

02) Variables that influence library behavior, appears in SADMIN Code Section of your code
We recommend, to set these variables prior to calling 'sa.sadm_start()'.

[001]	sa.ver	Program version number	03.28.00
[002]	sa.desc	Program description	Short description of program
[003]	sa.root_only	Program can only be run by 'root'	False
[004]	sa.server_only	Program can only run on SADMIN server	False
[005]	sa.sadmgrp_only	Run only if part of SADMIN Grp or root	False
[006]	sa.multiple_exec	Can run more than 1 copy simultaneously	False
[007]	sa.quiet	Quiet False=ErrNo+Mess True=OnlyErrNo	False
[008]	sa.use_rch	Save execution data to '.rch' file	False
[009]	sa.log_type	Send Output to [S]creen [L]og [B]oth	B
[010]	sa.log_append	True=Append to log, False=New log	False
[011]	sa.log_header	Generate header in log	True
[012]	sa.log_footer	Generate footer in log	True
[013]	sa.db_used	Need Access to SADMIN Database ?	True
[014]	sa.db_name	Database Name (Default:'sadmin')	sadmin
[015]	sa.sadm_pid_timeout	PID file default TimeToLive (Sec)	14400 sec
[016]	sa.sadm_lock_timeout	Maximun nb. sec. a host can be lock	7200 sec
[017]	sa.sadm_max_rchline	Trim rch file to this max. of lines	30 lines
[018]	sa.sadm_max_logline	Trim log to this maximum of lines	500 lines
[019]	sa.sadm_mail_addr	SADMIN Administrator Email(s)	brucetalbot95@gmail.com
[020]	sa.sadm_alert_type	0=NoMail 1=OnError 3=OnSuccess 4=All	1
[021]	sa.sadm_alert_repeat	0=AlertOnce or Sec. before alert repeat	0 sec
[022]	sa.sadm_alert_group	Error Group (Default Group)	default
[023]	sa.sadm_warning_group	Warning Group Name	warning
[024]	sa.sadm_info_group	Info Group Name	info

02) These variables are set by the Library and are available for you to use.
But shouldn't have to change.

[025]	sa.pn	Pgm. name with extension	sadmlib_std_demo.py
[026]	sa.inst	Pgm. name without extension	sadmlib_std_demo
[027]	sa.username	Current user name	root
[028]	sa.hostname	Current Host Name	sherlock
[029]	sa.pid	Current Process ID	2730214
[030]	sa.errno	Error No. returned by function called	0
[031]	sa.errmsg	Error Mess. returned by function called	
[032]	sa.db_conn	Database connector	<pymysql.connections.Connection object a

t 0x7fa1f32aa490>		
[033] sa.db_cur	Database cursor	<pymysql.cursors.DictCursor object at 0x7fa1f32aa430>
[034] sa.db_name	Database name	sadmin
[035] debug	Debug Level 0-9 (Incr. Verbose)	0
[036] exit_code	Script Exit Return Code	0
[037] current_time	Current date & Time	2026.07.10 20:08:27
[038] cmd_ssh_full	SSH cmd to client	/usr/bin/ssh -qnp 32

03) Python Library - Overview of the 'start()' and 'stop()' functions.

Function 'start()'

The 'start()' function basically initialize the SADMIN environment.

When you call 'start()', it will only come back to caller, only if everything went OK, Otherwise, it will advise the user of the error and exit(1).

- If 'db_used' is True, it will create a connection to the Database and return a connection (db_conn) and cursor (db_cur) objects.
Example: '(db_conn,db_cur) = sa.start()'
- If 'sa.db_used' is False, then it will return None for both 'sa_conn' and 'sadb_cur'.
Example: 'sa.start()'

Here a summary of the different things the 'start()' function does :

- 1) Make sure \$SADMIN directories and sub-directories exist and have proper permissions.
- 2) If 'log_append' is True, it open standard/error log in append mode, else in write mode.
- 3) If 'log_header' is True, write the log header.
- 4) If 'root_only' is True and the current user is not root, show error message and exit(1).
- 5) If 'server_only' is True and you are not on the SADMIN server, show error message and exit(1).
- 6) If 'sadmgrp_only' is 'True', then current user must be part of the SADMIN group (unless root).
- 7) If 'db_used' is True, open connection to database and return connection and cursor objects.
If not on SADMIN system, show error message and exit(1).
- 8) If lock_status() is True and running on the SADMIN server, issue message and exit(1).
If the creation of the lock was done more than 'lock_timeout' seconds, the lock file is remove.
- 9) If PID file 'pid_file' exist and 'sa.multiple_exec' is True, continue normal execution.
If PID file exist and execution time (sec) is less than the 'pid_timeout', show error mess. and exit(1).
If PID file exist and execution time (sec) exceed the 'pid_timeout' a new 'pid_file' is created and execution is resume.
- 10) If 'use_rch' is True, add a line in 'rch' file with code '2' and starting time of your program.

Function 'stop()'

This function accept one parameter, either a 0 for Success or a 1 for Error.
This should be the one of the last function called at the end of your program.

What this function does:

- 1) Calculate execution Time.
- 2) If 'use_rch = True', update the rch file (End Time & Elapse Time ...).
- 3) Validate & determine the alert group name (sadm_alert_group) use in rch file.
- 4) If 'db_used' is True and we are on the SADMIN server, close the database connection.
- 5) If the error log file (.elog) is empty, delete it.
- 6) Delete the PID file of the script (pid_file).
- 7) If 'log_footer' is True, write the log footer.
- 8) Close log and error log files.
- 9) If 'sa.max_logline' is not zero, trim the log according to user choice in 'sa.max_logline'.
- 10) If 'sa.max_rchline' is not zero, trim the 'rch' according to user choice in 'sa.max_rchline'.
- 11) Set permission and owner/group to log and rch files.
- 12) If on the SADMIN server, then rch and log are immediatly web central directory.

4) SADMIN configuration file content - List of variables available to developers.

----- General Section -----

[001]	sa.sadm_server	SADMIN server name (FQDN)	sherlock.maison.ca
[002]	sa.sadm_host_type	SADMIN [C]lient or [S]erver	S
[003]	sa.sadm_cie_name	Your Company name	The Batcave
[004]	sa.sadm_domain	Server creation default domain	maison.ca
[005]	sa.sadm_user	SADMIN User Name	sadmin
[006]	sa.sadm_group	SADMIN Group Name	sadmin
[007]	sa.sadm_pwd_random	Daily change password of '\$SADM_USER'	Y
[008]	sa.sadm_pid_timeout	PID file default TimeToLive (Sec)	14400 sec.
[009]	sa.sadm_lock_timeout	Maximun nb. sec. a host can be lock	7200 sec.

----- Database Section -----

[010]	sa.sadm_dbname	SADMIN Database Name	sadmin
[011]	sa.sadm_dbhost	SADMIN Database Host	localhost
[012]	sa.sadm_dbport	SADMIN Database Host TCP Port	3306
[013]	sa.sadm_ro_dbuser	SADMIN Database Read Only User	squery
[014]	sa.sadm_ro_dbpwd	SADMIN Database Read Only User Pwd	*Hidden*
[015]	sa.sadm_rw_dbuser	SADMIN Database Read/Write User	sadmin
[016]	sa.sadm_rw_dbpwd	SADMIN Database Read/Write User Pwd	*Hidden*

----- Web Interface Section -----

[017]	sa.sadm_www_user	User that Run the Web Server	apache
[018]	sa.sadm_www_group	Group that Run the Web Server	apache
[019]	sa.sadm_monitor_update_interval	Monitor web page refresh rate (Sec)	60 Seconds
[020]	sa.sadm_monitor_recent_count	Monitor web page nb. recent scripts	5

[021]	sa.sadm_monitor_recent_exclude	Monitor web page exclude list	sadm_nmon_watcher,sysmon_script_test
----- Email Section -----			
[022]	sa.sadm_mail_addr	SADMIN SysAdmin Email(s)	brucetalbot95@gmail.com
[023]	sa.sadm_smtp_server	Your internet smtp server	smtp.gmail.com
[024]	sa.sadm_smtp_port	Your internet smtp server port	587
[025]	sa.sadm_smtp_sender	Your internet smtp email address	brucetalbot95@gmail.com
[026]	sa.sadm_gmpw	Your internet smtp email password	*Hidden*
[027]	sa.sadm_email_startup	Send email to SysAdmin on startup	Y
[028]	sa.sadm_email_shutdown	Send email to SysAdmin on shutdown	Y
----- Monitoring Section -----			
[029]	sa.sadm_alert_type	0=NoAlert 1=OnError 3=OnSuccess 4=All	1
[030]	sa.sadm_alert_group	Error Group Name (Default Group)	default
[031]	sa.sadm_warning_group	Warning Group Name	warning
[032]	sa.sadm_info_group	Info Group Name	info
[033]	sa.sadm_ssh_port	Default SSH Port to talk with client	32
[034]	sa.sadm_textbelt_key	TextBelt.com API Key	*Hidden*
[035]	sa.sadm_textbelt_url	TextBelt.com API URL	https://textbelt.com/text
[036]	sa.sadm_ntfy_email	Email related to NTFY	duplessis.jacques@gmail.com
[037]	sa.sadm_ntfy_pwd	Password related to NTFY	5ne46yJFqmQS24rU8e9
[038]	sa.sadm_ntfy_token	Token related to NTFY	tk_2a511lu2o46k5hdglt64h5u55wt0
[039]	sa.sadm_ntfy_topic	Topic use with NTFY	sadmdev
[040]	sa.sadm_ntfy_url	URL to send notification to NTFY	https://ntfy.sh
[041]	sa.sadm_ntfy_user	NTFY user name	duplessis.jacques@gmail.com
----- Files and Logs pruning Section -----			
[042]	sa.sadm_nmon_keepdays	Nb. of days to keep .nmon perf. file	10 days
[043]	sa.sadm_rch_keepdays	Nb. days to keep unmodified .rch file	35 days
[044]	sa.sadm_log_keepdays	Nb. days to keep unmodified .log file	35 days
[045]	sa.sadm_max_rchline	Trim rch file to this max. of lines	30 lines
[046]	sa.sadm_max_logline	Trim log to this maximum of lines	500 lines
[047]	sa.sadm_days_history	Days to keep alert in History file	100 days
[048]	sa.sadm_max_arc_line	Max. lines in alert in History file	1000 lines
----- Subnet Network Scanner Section -----			
[049]	sa.sadm_network1	Network/Netmask 1 inv. IP/Name/Mac	192.168.1.0/24
[050]	sa.sadm_network2	Network/Netmask 2 inv. IP/Name/Mac	None
[051]	sa.sadm_network3	Network/Netmask 3 inv. IP/Name/Mac	None
[052]	sa.sadm_network4	Network/Netmask 4 inv. IP/Name/Mac	None
[053]	sa.sadm_network5	Network/Netmask 5 inv. IP/Name/Mac	None
----- ReaR Backup Section -----			
[054]	sa.sadm_rear_nfs_server	ReaR NFS Server IP or Name	lestrade.maison.ca
[055]	sa.sadm_rear_nfs_server_ver	ReaR NFS Server Version	3
[056]	sa.sadm_rear_nfs_mount_point	ReaR NFS Mount Point	/backup/backup_rear
[057]	sa.sadm_rear_backup_to_keep	ReaR NFS Backup - Nb. to keep	2 copies
[058]	sa.sadm_rear_backup_dif	Alert if pct between cur. & prev. size	25 %

[059]	sa.sadm_rear_backup_interval	Alert if no backup for more than X days	14 days
[060]	sa.sadm_rear_backup_script	ReaR backup script name	sadm_rear_backup.sh
[061]	sa.sadm_rear_del_failed_backup	Del backup if failed integrity check	Y
[062]	sa.sadm_rear_backup_concurrent	Number of Concurrent rear backup	1
[063]	sa.sadm_rear_batch_mode	Run backup in batch mode (y/N)	N
[064]	sa.sadm_rear_batch_startup_time	Batch startup time	00:10
[065]	sa.sadm_rear_batch_day2run	0=AnyDay, 1=Su, 2=Mo, 3=We, 4=Tu, 6=Fr, 7=Sa	"2,7"
[066]	sa.sadm_rear_batch_mth2run	0=AnyMonth or [1,2,,12] Month to run	"0"
[067]	sa.sadm_rear_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

----- Backup Section -----

[068]	sa.sadm_backup_nfs_server	NFS Backup IP or Server Name	batnas.maison.ca
[069]	sa.sadm_backup_nfs_server_ver	NFS Backup Server Version	3
[070]	sa.sadm_backup_nfs_mount_point	NFS Backup Mount Point	/volume1/backup/backup_sadmin
[071]	sa.sadm_backup_dif	Alert if pct between cur. & prev. size	40 %
[072]	sa.sadm_backup_interval	Alert if no backup for more than X days	1 days
[073]	sa.sadm_daily_backup_to_keep	Daily Backup to Keep	2 copies
[074]	sa.sadm_weekly_backup_to_keep	Weekly Backup to Keep	1 copies
[075]	sa.sadm_monthly_backup_to_keep	Monthly Backup to Keep	1 copies
[076]	sa.sadm_yearly_backup_to_keep	Yearly Backup to keep	1 copies
[077]	sa.sadm_weekly_backup_day	Weekly Backup Day (1=Mon, 7=Sun)	5
[078]	sa.sadm_monthly_backup_date	Monthly Backup Date (1-28)	1
[079]	sa.sadm_yearly_backup_month	Yearly Backup Month (1-12)	12
[080]	sa.sadm_yearly_backup_date	Yearly Backup Date (1-31)	31
[081]	sa.sadm_backup_concurrent	Number of Concurrent Backup	1
[082]	sa.sadm_backup_batch_mode	Run backup in batch mode (y/N)	N
[083]	sa.sadm_backup_batch_startup_time	Batch startup time	00:05
[084]	sa.sadm_backup_batch_day2run	0=AnyDay, 1=Su, 2=Mo, 3=We, 4=Tu, 6=Fr, 7=Sa	"2,7"
[085]	sa.sadm_backup_batch_mth2run	0=AnyMonth or [1,2,,12] Month to run	"0"
[086]	sa.sadm_backup_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

----- Virtual Machines Export Section -----

[087]	sa.sadm_vm_export_nfs_server	NFS Export Server	batnas.maison.ca
[088]	sa.sadm_vm_export_nfs_server_ver	NFS Export Server Version	3
[089]	sa.sadm_vm_export_mount_point	NFS Export Mount Point	/volume1/backup/backup_vm/virtualbox_exports
[090]	sa.sadm_vm_export_dif	Alert if pct between cur. & prev. size	25 %
[091]	sa.sadm_vm_export_to_keep	Nb. of export to keep	2 copies
[092]	sa.sadm_vm_export_interval	Alert if no export for more than X days	21 days
[093]	sa.sadm_vm_export_alert	Issue an alert if interval reached	Y
[094]	sa.sadm_vm_user	User part of 'vboxusers' user group	jacques
[095]	sa.sadm_vm_stop_timeout	Max. seconds given for acpi shutdown	60 seconds
[096]	sa.sadm_vm_start_interval	Sec. to sleep between each VM start	30 seconds
[097]	sa.sadm_vm_export_concurrent	Number of Concurrent Export	1
[098]	sa.sadm_vm_export_batch_mode	Run export in batch mode (y/N)	N
[099]	sa.sadm_vm_export_batch_start_time	Start time for launch batch export	00:01
[100]	sa.sadm_vm_export_batch_day2run	0=AnyDay, 1=Su, 2=Mo, 3=We, 4=Tu, 6=Fr, 7=Sa	"2,7"
[101]	sa.sadm_vm_export_batch_mth2run	0=AnyMonth or [1,2,,12] Month to run	"0"

[102]	sa.sadm_vm_export_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"
----- System Update Section -----			
[103]	sa.sadm_osupdate_script	System Update Script Name	sadm_osupdate.sh
[104]	sa.sadm_osupdate_interval	Alert if no update for more than X days	15 days
[105]	sa.sadm_osupdate_autoremove	Auto remove 'orphaned' packages.	N
[106]	sa.sadm_osupdate_flatpak	Update Flatpak packages	N
[107]	sa.sadm_osupdate_snap	Update Snap packages	N
[108]	sa.sadm_osupdate_reboot_needed	Reboot after each update	N
[109]	sa.sadm_osupdate_reboot_time	Sec. given to bring up O/S & app.	240 seconds
[110]	sa.sadm_osupdate_lock	Lock system during the update	Y
[111]	sa.sadm_osupdate_concurrent	Number of Concurrent System Updates	1
[112]	sa.sadm_osupdate_batch_mode	Run O/S Update in Batch Mode (y/N)	N
[113]	sa.sadm_osupdate_batch_start_time	Batch Update Start Time	00:05
[114]	sa.sadm_osupdate_batch_day2run	0=AnyDay, 1=Su, 2=Mo, 3=We, 4=Tu, 6=Fr, 7=Sa	"2,7"
[115]	sa.sadm_osupdate_batch_mth2run	0=AnyMonth or [1,2,,,12] Month to run	"0"
[116]	sa.sadm_osupdate_batch_date2run	0=AnyDate, Date to run [1,2...27,28]	"0"

5) Directories variables available to Developers

[001]	sa.dir_base	SADMIN Root Directory	/opt/sadmin
[002]	sa.dir_bin	SADMIN Scripts Directory	/opt/sadmin/bin
[003]	sa.dir_tmp	System Temporary file(s) Directory	/opt/sadmin/tmp
[004]	sa.dir_lib	SADMIN Shell & Python Library Dir.	/opt/sadmin/lib
[005]	sa.dir_log	SADMIN Script Log Directory	/opt/sadmin/log
[006]	sa.dir_cfg	SADMIN Configuration Directory	/opt/sadmin/cfg
[007]	sa.dir_sys	System Startup/Shutdown Script Dir.	/opt/sadmin/sys
[008]	sa.dir_doc	SADMIN Documentation Directory	/opt/sadmin/doc
[009]	sa.dir_pkg	SADMIN Packages Directory	/opt/sadmin/pkg
[010]	sa.dir_dat	System Data Directory	/opt/sadmin/dat
[011]	sa.dir_nmon	System 'nmon' - Data Collected Dir.	/opt/sadmin/dat/nmon
[012]	sa.dir_dr	System Disaster Recovery Info Dir.	/opt/sadmin/dat/dr
[013]	sa.dir_rch	System Return Code History Dir.	/opt/sadmin/dat/rch
[014]	sa.dir_net	System Network Information Dir.	/opt/sadmin/dat/net
[015]	sa.dir_rpt	System Monitor Report Directory	/opt/sadmin/dat/rpt
[016]	sa.dir_dbb	SADMIN Database Backup Directory	/opt/sadmin/dat/dbb
[017]	sa.dir_setup	SADMIN Setup Directory.	/opt/sadmin/setup
[018]	sa.dir_usr	User root directory	/opt/sadmin/usr
[019]	sa.dir_usr_bin	User Scripts Directory	/opt/sadmin/usr/bin
[020]	sa.dir_usr_lib	User Library Directory	/opt/sadmin/usr/lib
[021]	sa.dir_usr_doc	User Documentation Dir.	/opt/sadmin/usr/doc
[022]	sa.dir_usr_mon	User System Monitor Scripts	/opt/sadmin/usr/mon
[023]	sa.dir_www	SADMIN Web Site Root Directory	/opt/sadmin/www
[024]	sa.dir_www_doc	SADMIN Web Site Documentation Dir.	/opt/sadmin/www/doc
[025]	sa.dir_www_dat	SADMIN Web Site Global Data Dir.	/opt/sadmin/www/dat

[026]	sa.dir_www_lib	SADMIN Web Site PHP Library Dir.	/opt/sadmin/www/lib
[027]	sa.dir_www_tmp	SADMIN Web Temp Working Dir.	/opt/sadmin/www/tmp
[028]	sa.dir_www_perf	SADMIN Web Performance Graph Dir.	/opt/sadmin/www/tmp/perf

6) SADMIN File names use and set by the Python library, for you to use

----- Available File Name Aliases Variables -----

[001]	sa.pid_file	Current script PID file	/opt/sadmin/tmp/sadmlib_std_demo.pid
[002]	sa.cfg_file	SADMIN Main Configuration file	/opt/sadmin/cfg/sadmin.cfg
[003]	sa.cfg_hidden	SADMIN configuration template file	/opt/sadmin/cfg/.sadmin.cfg
[004]	sa.alert_file	Alert Group definition file	/opt/sadmin/cfg/alert_group.cfg
[005]	sa.alert_init	Alert Group definition template file	/opt/sadmin/cfg/.alert_group.cfg
[006]	sa.alert_hist	Alert History file	/opt/sadmin/cfg/alert_history.txt
[007]	sa.alert_hini	Alert History template file	/opt/sadmin/cfg/.alert_history.txt
[008]	sa.tmp_file1	User usable Temp Work File 1	/opt/sadmin/tmp/tmp_file_1.2730214
[009]	sa.tmp_file2	User usable Temp Work File 2	/opt/sadmin/tmp/tmp_file_2.2730214
[010]	sa.tmp_file3	User usable Temp Work File 3	/opt/sadmin/tmp/tmp_file_3.2730214
[011]	sa.log_file	Log File Name	/opt/sadmin/log/sherlock_sadmlib_std_dem
o.log			
[012]	sa.rch_file	Return Code History File Name	/opt/sadmin/dat/rch/sherlock_sadmlib_std
_demo.rch			
[013]	sa.dbpass_file	SADMIN Database User Password File	/opt/sadmin/cfg/.dbpass
[014]	sa.rpt_file	[SYS]tem [MON]itor report file	/opt/sadmin/dat/rpt/sherlock.rpt
[015]	sa.backup_list	Backup file list	/opt/sadmin/cfg/backup_list.txt
[016]	sa.backup_list_init	Backup file template file	/opt/sadmin/cfg/.backup_list.txt
[017]	sa.backup_exclude	Backup exclude list file name	/opt/sadmin/cfg/backup_exclude.txt
[018]	sa.backup_exclude_init	Backup exclude template file	/opt/sadmin/cfg/.backup_exclude.txt

7) Command Path set and use by SADMIN Python Library for more security.

[001]	sa.cmd_dmidcode	Cmd. 'dmidecode', Get model & type	/usr/sbin/dmidecode
[002]	sa.cmd_bc	Cmd. 'bc', Do some Math.	/usr/bin/bc
[003]	sa.cmd_fdisk	Cmd. 'fdisk', Get Partition Info	/usr/sbin/fdisk
[004]	sa.cmd_which	Cmd. 'which', Get Command location	/usr/bin/which
[005]	sa.cmd_perl	Cmd. 'perl', epoch time Calc.	/usr/bin/perl
[006]	sa.cmd_mutt	Cmd. 'mutt', Used to Send Email	/usr/bin/mutt
[007]	sa.cmd_curl	Cmd. 'curl', To send alert to Slack	/usr/bin/curl
[008]	sa.cmd_lscpu	Cmd. 'lscpu', Socket & thread info	/usr/bin/lscpu
[009]	sa.cmd_nmon	Cmd. 'nmon', Collect Perf Statistic	/usr/bin/nmon
[010]	sa.cmd_parted	Cmd. 'parted', Get Disk Real Size	/usr/sbin/parted
[011]	sa.cmd_ethtool	Cmd. 'ethtool', Get System IP Info	/usr/sbin/ethtool
[012]	sa.cmd_rrdtool	Cmd. 'rrdtool' to produce graph	/usr/bin/rrdtool


```
[013] sa.cmd_lsb_release      Cmd. 'lsb_release' to get dist. info      /usr/bin/lsb_release
[014] sa.cmd_inxi             Cmd. 'inxi' binary location                  /usr/bin/inxi
[015] sa.cmd_ssh              Cmd. 'ssh', SSH to SADMIN client              /usr/bin/ssh
```

8) Database Information

```
[001] sa.db_used              Program need to access Database ?      True
```

Show SADMIN Tables:

Tables_in_sadmin
pma__bookmark
pma__central_columns
pma__column_info
pma__designer_settings
pma__export_templates
pma__favorite
pma__history
pma__navigationhiding
pma__pdf_pages
pma__recent
pma__relation
pma__savedsearches
pma__table_coords
pma__table_info
pma__table_uiprefs
pma__tracking
pma__userconfig
pma__usergroups
pma__users
script
script_rch
server
server_category
server_group
server_network

Category Table:

Field	Type	Null	Key	Default	Extra
-------	------	------	-----	---------	-------

cat_id		int (11)	NO	PRI	NULL	auto_increment			
cat_code		varchar(10)	NO	UNI	NULL				
cat_desc		varchar(25)	NO		NULL				
cat_active		tinyint(1)	NO		1				
cat_date		timestamp	NO		0000-00-00 00:00:00	on update current_timestamp()			
cat_default		tinyint(1)	NO		0				
Table		Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed
Null	Index_type	Comment	Index_comment						
server_category		0	PRIMARY	1	cat_id	A	8	NULL	NULL
	BTREE								
server_category		0	key_cat_code	1	cat_code	A	8	NULL	NULL
	BTREE								
cat_id	cat_code	cat_desc	cat_active	cat_date	cat_default				
13	Laptop	Laptop Workstation	1	2023-10-26 12:58:43	0				
15	SBC	Raspberry Pi SBC	1	2023-10-21 11:25:49	0				
17	Desktop	Workstation	1	2023-10-21 11:23:25	0				
25	VM	Virtualize Systems	1	2023-12-31 11:21:58	0				
28	Dell	Dell Systems	1	2023-10-26 12:59:19	0				
29	HP	HP Systems	1	2023-10-26 12:59:38	0				
30	IBM	IBM Systems	1	2023-10-26 12:59:05	0				
31	Lenovo	Lenovo Systems	1	2023-10-26 12:59:54	0				
32	Server	Home Server	1	2023-12-31 11:21:35	1				

Group Table:

Field	Type	Null	Key	Default	Extra				
grp_id	int(11)	NO	PRI	NULL	auto_increment				
grp_code	varchar(10)	NO	UNI	NULL					
grp_desc	varchar(25)	NO		NULL					
grp_active	tinyint(1)	NO		1					
grp_date	timestamp	NO		current_timestamp()	on update current_timestamp()				
grp_default	tinyint(1)	NO		0					
Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Nu

11	Index_type	Comment	Index_comment										
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													
	server_group		0	PRIMARY		1	grp_id	A		9	NULL	NULL	
	BTREE												
	server_group		0	idx_grp_code		1	grp_code	A		9	NULL	NULL	
	BTREE												
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													
	grp_id		grp_code		grp_desc		grp_active		grp_date		grp_default		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													
	4		Retired		System not in use		1		2023-04-05 11:26:50		0		
	6		Regular		Application system		1		2023-10-21 11:15:00		0		
	9		Dev		Development Systems Group		1		2023-10-21 11:16:20		0		
	10		Prod		Production Systems Group		1		2023-10-21 11:16:34		1		
	13		Int		Integration Systems Group		1		2023-10-21 11:20:19		0		
	16		Acc		Acceptance Systems Group		1		2023-10-22 16:38:23		0		
	17		Poc		Proof of concept Systems		1		2023-10-21 15:06:48		0		
	19		PreProd		Pre-Production Group		1		2023-10-22 16:39:19		0		
	20		Temp		Temporary Systems Group		1		2023-10-26 11:44:28		0		
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+													

Server Table:

Field	Type	Null	Key	Default	Extra
srv_id	int(11)	NO	PRI	NULL	auto_increment
srv_name	varchar(15)	NO	UNI	NULL	
srv_domain	varchar(30)	NO		NULL	
srv_desc	varchar(30)	NO		NULL	
srv_tag	varchar(15)	YES		NULL	
srv_note	varchar(30)	YES		NULL	
srv_active	tinyint(1)	NO		1	
srv_sporadic	tinyint(1)	NO		0	
srv_monitor	tinyint(1)	NO		1	
srv_alert_group	varchar(15)	NO		default	
srv_graph	tinyint(1)	NO		1	
srv_cat	varchar(10)	NO		NULL	
srv_group	varchar(10)	NO		NULL	
srv_vm	tinyint(1)	NO		0	
srv_osname	varchar(15)	YES		NULL	
srv_ostype	varchar(10)	NO		linux	
srv_oscodename	varchar(15)	YES		NULL	
srv_osversion	varchar(12)	YES		NULL	
srv_osver_major	varchar(10)	YES		NULL	
srv_date_creation	datetime	YES		0000-00-00 00:00:00	

[illegible]

[illegible]

[illegible]

Script table:

Field	Type	Null	Key	Default	Extra
scr_id	mediumint(8) unsigned	NO	PRI	NULL	auto_increment
scr_name	varchar(40)	YES	UNI	NULL	
scr_desc	varchar(80)	YES		NULL	
scr_version	varchar(10)	NO		NULL	
scr_alert_grp	varchar(15)	YES		NULL	
scr_alert_type	smallint(6)	NO		1	
scr_sysadm_email	varchar(50)	NO		NULL	
scr_multiple_exec	tinyint(1)	NO		0	
scr_use_rch	tinyint(1)	NO		1	
scr_log_type	varchar(1)	NO		B	
scr_log_append	tinyint(1)	NO		0	
scr_log_header	tinyint(1)	NO		1	
scr_log_footer	tinyint(1)	NO		1	
scr_active	tinyint(1)	NO		1	
scr_max_logline	int(11)	NO		500	
scr_max_rchline	int(11)	NO		35	
scr_pid_timeout	int(11)	NO		7200	
src_max_inactive_days	int(3)	NO		31	
scr_root_exec	tinyint(4)	NO		0	
scr_run_sadmin	tinyint(4)	NO		0	
scr_os	varchar(10)	NO		any	
scr_last_activity	datetime	NO		current_timestamp()	
scr_severity	smallint(6)	NO		1	
scr_exclude	set('0','1')	NO		0	
scr_log_keepdays	smallint(6)	NO		0	
scr_rch_keepdays	smallint(6)	NO		0	
scr_avg_exec	int(11)	YES		0	
scr_debug	tinyint(1)	NO		0	
scr_grp_warning	varchar(15)	NO		NULL	
scr_grp_info	varchar(15)	NO		NULL	
scr_grp_error	varchar(15)	NO		NULL	
scr_lock_timeout	int(11)	NO		3600	

Table index_type	Non_unique Comment	Key_name Index_comment	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null	I
---------------------	-----------------------	---------------------------	--------------	-------------	-----------	-------------	----------	--------	------	---

script	0	PRIMARY	1	scr_id	A		0	NULL	NULL		B
TREE											
script	0	scr_name_key	1	scr_name	A		0	NULL	NULL	YES	B
TREE											

Script_rch:

Field	Type	Null	Key	Default	Extra
scrdet_host	varchar(15)	NO	PRI	NULL	
scrdet_name	varchar(40)	NO	PRI	NULL	
srcdet_start	timestamp	NO	PRI	0000-00-00 00:00:00	
srcdet_end	timestamp	YES		NULL	
scrdet_status	tinyint(2)	NO		0	
scrdet_active	int(1)	NO		1	
srcdet_elapse	time	YES		NULL	
scrdet_alertcode	tinyint(3) unsigned	YES		NULL	
scrdet_alertgroup	varchar(30)	YES		NULL	

Table	Non_unique	Key_name	Seq_in_index	Column_name	Collation	Cardinality	Sub_part	Packed	Null
Index_type	Comment	Index_comment							
script_rch	0	PRIMARY	1	scrdet_host	A	0	NULL	NULL	
BTREE									
script_rch	0	PRIMARY	2	scrdet_name	A	0	NULL	NULL	
BTREE									
script_rch	0	PRIMARY	3	srcdet_start	A	0	NULL	NULL	
BTREE									

Closing Database db_connection

Cursor closed successfully.

Successfully closed database connection.

Script exit code is 0 (Success) and execution time is 00:00:00
History file \$SADMIN/dat/rch/sherlock_sadmlib_std_demo.rch trim to 50 lines.
Send alert only when it terminate with error.

Log file \$SADMIN/log/sherlock_sadmlib_std_demo.log trim to 500 lines.
Fri 10 Jul 2026 20:08:27 - End of sadmlib_std_demo.py
